INCH-POUND

MIL-C-50652/3A (AR)
09 April 1993
SUPERSEDING
MIL-C-50652/3(MU)
15 April 1971

MILITARY SPECIFICATION SHEET

CIRCUIT **CARD ASSEMBLY:** 10548119

(DECODER, RCFD)

This specification is approved for use by the U.S. Army - Armament, Munitions and Chemical Command, and is available for by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The detail requirements of this specification sheet together with the general requirements of MIL-C-50652 comprise the **complete requirements for the Range Control Fine Delay Decoder (RCFD Decoder) used in the Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2. The RFCD Decoder forms part of a binary to decimal decoder for 1.25 meter range increments, provides a single output called** the **E/L coarse output from the RCFD high and low data, and has circuits for detecting minimum and** maximum **range rates and velocity of less than 10 meters per second.**

2. **APPLICABLE DOCUMENTS**

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks- The following specifications, standards, and handbooks form a part of this document to the **extent specified herein. Unless** otherwise specified, the issues of these documents are be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-50652 Circuit **card Assemblies,** General Specification for

, ts **(recommendation ? additions, deletions) and any pertinent data which may be of use in improving this document sjpi;d be addressed to: Commander, U.S. ARDEC, ATTN:SMCAR-BAC-s, Picatinny,Arsenal, I** New Jersey 07806-5000 by using the self-addressed Standardbduaization Document **Improvement Proposal (DD Form 1426)**

AMSC N/A **FSC 1285**

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:31Z
Check the source to verify that this is the current version before use.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548119 - Decoder, RCFD

SPECIAL PACKAGING INSTRUCTION

PS10548119 Packaging of Decoder, RCFD

(Copies of other Government documents, drawings, and publications, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, shall supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.2 General specifications. The assembly shall meet the requirements of MIL-C-50652.

3.2.1 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10548119, all associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Environmental service conditions. AS specified in MIL-C-50652.

3.4. Electrical. Each conductor shall be electrically continuous as specfied on the schematic diagram of the card assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

3.5 Performance. For the requirements of 3.5.3 through 3.5.5, logic input and output levels shall be as fellows:

Logic o: 0.0 + 0.5 volt direct currect (Vdc) level.
Logic 1: + 3.5 ± 1.0 Vdc level.

3.5.1 Input signals. For all requirements except 3.5.5.2 the input signals to the RCFD decoder shall be T1, T3, T1 and T3 as shown in Figure 1. For requirement of 3.5.5.2 a T1 signal as shown in Figure 1 will be alternately applied to P1-41 and 11. A positive pulse shall be a change in direct current (de) level from 0.0 + 0.5 volt to + 3.5 ± 1.0 V and return to 0 V. A negative pulse shall be a change in dc level from + 3.5 ± 1.0 V to 0.0 + 0.5 V and return to + 3.5 V.

3.5.2 **Input power.** The input power applied between terminals P1-30, 31 and 32 (+5 V) and terminals Pl-24, 25 and 26 (+5 V return) shall be 5.0 ± 0.3 Vdc, at 300 milliamperes (mA), regulated; with an alternating current (at) ripple less than 50 millivolts (mV) peak-to-peak.

3.5.3 Velocity output signals. With input logic levels applied sequentially as specified in Table I, the ~~R Max~~, ECM R <+ 5 meters per second (m/s) and R Min output signals at Pl-12, Pl-14, and Pl-38 respectively shall be as specified in Table 1. The output pulses shall have the same characteristics as shown in Figure 1.

TABLE I. WAVEFORM OBSERVATIONS,

INPUT TERMINALS					OUTPUT SIGNAL CHARACTERISTICS		
STEP	16	22	50	1	14	12	38
1	1	0	1	1	T3 pulse	T1 pulse	Logic 1
2	0	1	0	1	T1 pulse	Logic 1	T3 pulse
3	1	0	1	0	Logic 1	-------	--------
4	0	1	0	0	Logic 1	------ --	-0------

3.5.4 RCFD decoder output. With an input logic O applied to P1-56, P1-54, P1-44, and P1-52 and signal T1, as shown in Figure 1, applied to P1-42, 46, 49 and 51; the T1 signal as shown In Figure 1 shall be present at the following terminals, P1-3, P1-7, P1-13, Pl-17, P1-5, P1-9, P1-15, P1-21, P1-23, P1-37, P1-43, P1-47, P1-35, Pl-39, P1-45, and P1-48. With an input logic 1 applied to P1-56, Pl-54, P1-44, and P1-52 the outputs above shall all be logic O.

3.5.5 E/L trigger output.

3.5.5.1 E/L output. With P1-55 floating, signal T1 & T3 as shown in Figure 1 shall be present at P1-19.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

3.5.5.2 **E/L** delay differential. With signals T1 and T3 as shown in Figure 1 being alternately applied to P1-11, and 41 **(T1 to P1-11 and T3 to P1-41; then T1 to P1-41 and T3 to P1-11) and P1-55 floating, the output signal at P1-19 shall be T1 and T3 as shown in Figure 1. As the input signals are alternately applied the delay differential of the output signal (From the leading edge of the input signal to the leading edge of each alternative T1 pulse) shall** not **show a delay differential greater than 1.0 nanosecond (NS).**

3.6 First Article. As specified in MIL-C-50652.

3.7 Soldering. As specified in MIL-C-50652.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. The quality assurance provisions of MIL-C-50652 shall apply to the assembly covered by this detail specification **sheet of the circuit card assembly.**

4.2 First article. As specified In MIL-C-50652.

4.3 Inspection provisions. As specified in MIL-C-50652.

4.4 Special sampling. As specified in MIL-C-50652.

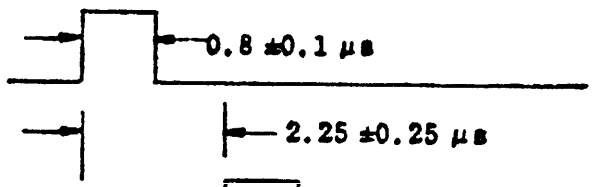
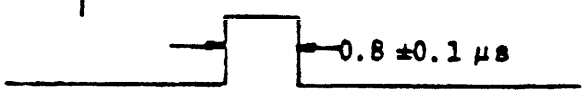
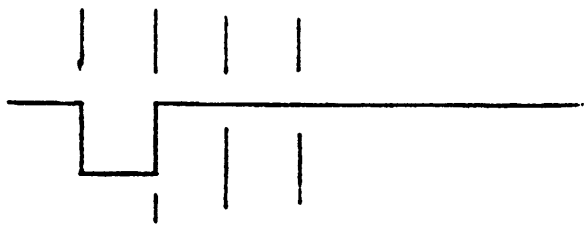
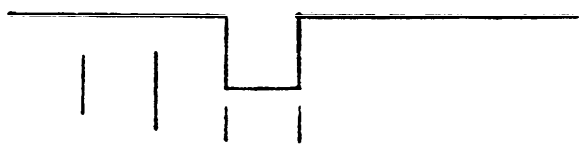
4.5 Inspection equipment. As specified in MIL-C-50652 and **4.5.2.3.**

4.5.1 Government-furnished inspection equipment. As specified in **MIL-C-50652.**

4.5.2 Contractor-furnished inspection equipment. As specified in **MIL-C-50652.**

4.5.3 Test equipment. **In conjunction** with 4.5.1 and **4.5.2 the** test equipment" listed in Table **II shall be utilized in the performance** of **the applicable tests as specified in 4.6.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

SIGNAL NAME	WAVESHAPE OR CHARACTERISTIC	TERMINALS
T1		P1-41
T3		P1-11
$\overline{T1}$		P1-18, P1-42, P1-46, P1-49, & P1-51
$\overline{T3}$		P1-34
T1 & T3		P1-19

NOTE : 1. All signals at a 20-kHz PRF.

FIGURE 1. INPUT-OUTPUT SIGNAL CHARACTERISTICS.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:31Z
Check the source to verify that this is the current version before use.

TABLE II. TEST EQUIPMENT LIST.

ITEM NUMBER	DESCRIPTION
1. Power supply	5.0 ± 0.3 Vdc, regulated. Ac ripple: less than 50 mV peak-to-peak Capacity: 300 mA.
2. Pulse generator	Frequency: 20 KHz minimum Duty factor: 10% minimum Amplitude: minimum of 4 volts into a SO-ohm load. Pulse: Rise fall times less than 0.1 us.
3. Pulse generator	Same as item 2.
4. Oscilloscope	Rise time of scope plus measuring probe 10 ns or less, sweep period 50 ns per division or less, input resistance greater than 8 megohms, input capacitance less than 25 uF.
5. Sampling oscilloscope	Rise time of scope plus measuring probe 100 ps or less, sweep period 1 ns per division or less; input resistance greater than 500 ohms, input capitance less than 10 pF.
6. Test fixture	As specified in Figure 2. See Table III for test fixture parts list.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

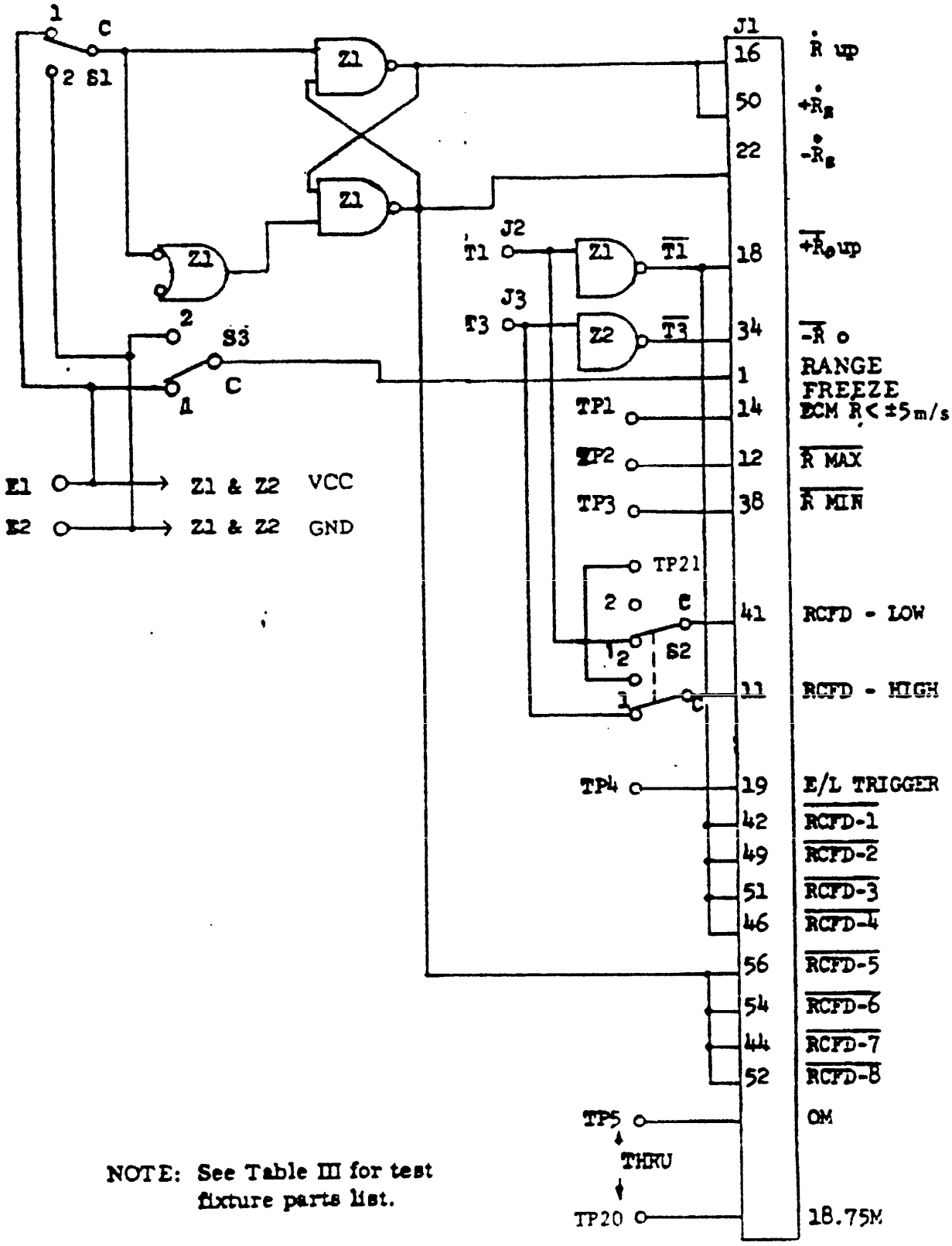


FIGURE 2. TEST FIXTURE SETUP

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

TABLE III. TEST FIXTURE PARTS LIST.

DESIGNATION	NOMENCLATURE AND DESCRIPTION
E1	Binding post, H. H. Smith 257-R or equivalent.
E2	Binding post, H. H. Smith 257-B or equivalent.
J1	Connector; any of the following or equivalent.
	(1) 5893232-1, Malco Mfg. Co., Inc. Chicago, Illinois.
	(2) 500946-01, National Connector Corp. Minneapolis, Mire.
	(3) 07-5422-0198-01, Elco Corp., Willow Grove, PA
J2	Connector, receptacle, UG-25B/U or equivalent.
J3	Same as J2.
S1 and S3	Switch SPDT.
S2	Switch DPDT.
TPI thru TP21	Tip jack, H. H. Smith 1501 or equivalent.
Z1 and 22	Integrated circuit, Quad 2-Input NAND SN5400 or equivalent.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:31Z
Check the source to verify that this is the current version before use.

4.6 Test methods and procedures.

4.6.1 Environmental. As specified in MIL-C-50652.

4.6.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and **visual examination for compliance with 3.4.**

4.6.3 Performance. The selected RCFD decoders shall be tested as follows: Set up equipment as shown in Figure 3. Use oscilloscope, item 4, for observation of all input and output signals as specified in 4.6.3.2 through 4.6.3.5.1. Use oscilloscope, item 5, for observation of output signals as specified in 4.6.3.5.2. When specified, a logic 0 shall be observed as a 0.0 + 0.5 Vdc level and a logic 1 as a +3.5 + 1.0 Vdc level. Switch S1 on test fixture, item 6, is used to control logic level input signals to the RCFD assembly. Switch S1, when in position 1, applies logic 0 to P1-16, 50 and 55, and logic 1 to Pi-22, 44, 52, 54, and 56. With S1 in position 2 all logic levels are complemented.

4.6.3.1 Input power. Using item 1, apply input power **as specified in 3.5.1.**

4.6.3.2 Input signals. Using pulse generators, items 2 and 3, set up and apply input signals T1 and T3 as shown in Figure 1, to **J2 and J3 on test fixture, item 6.**

4.6.3.3 Velocity output signals. At TP1, TP2, and TP3 on the test fixture, item 6, Observe the ECM R < ± sm/s, R MAX and R Min output signals as specified in 3.5.3.

4.6.3.4 RCFD decoder signals. With logic levels applied, as specified in 3.5.4. (Sl in position 2), at TP5 through TP20, observe output signal as specified in 3.5.4. with Sl in position 1, all outputs (TP5-TP20) shall be logic 0.

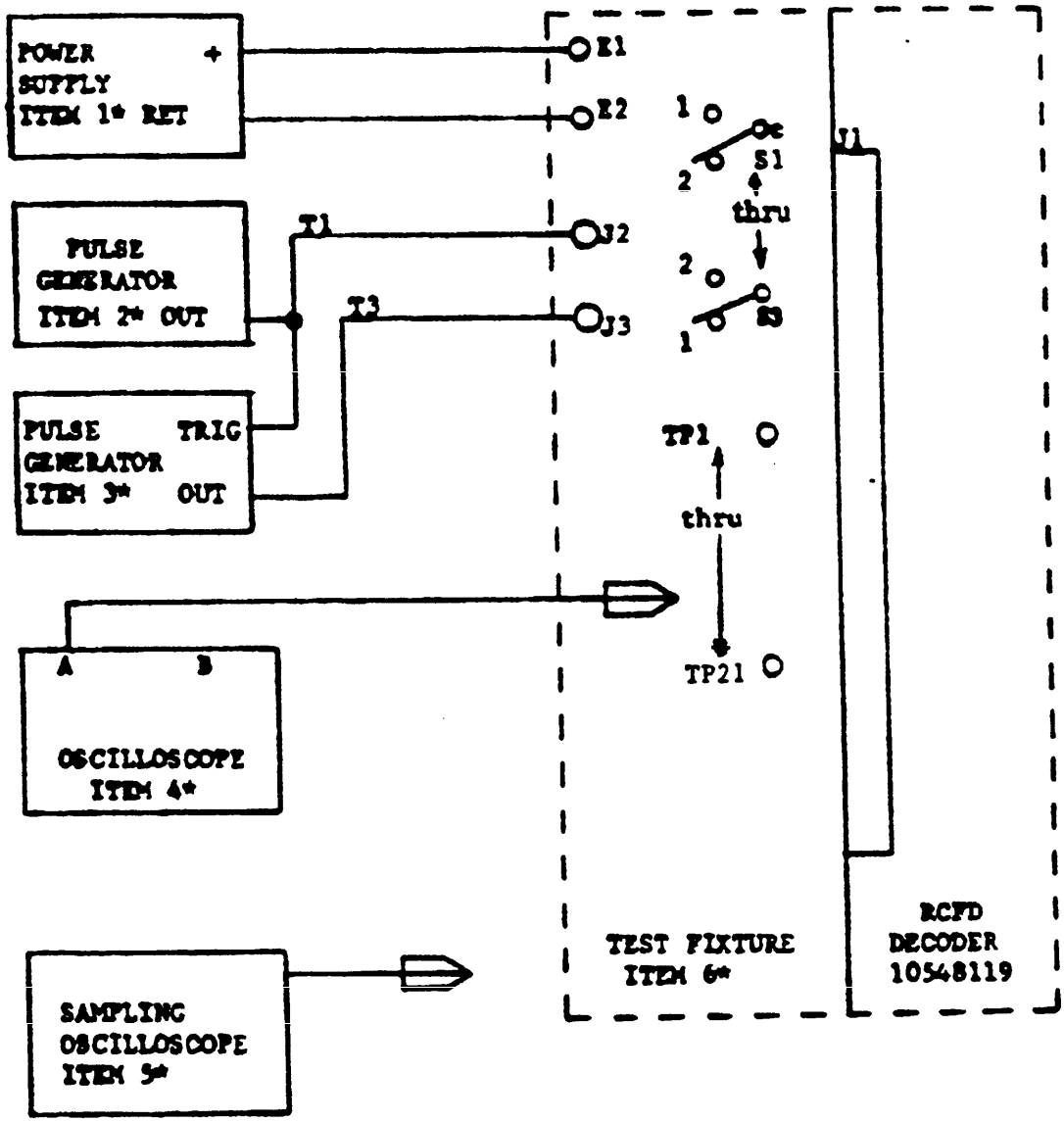
4.6.3.5 E/L trigger output.

4.6.3.5.1 E/L output. **With a logic 1 level applied as specified in 3.5.5.1, observe at** TP4 the output signals as specified in 3.5.5.1.

4.6.3.5.2 E/L delay differential. Apply signal T1 and T3 as specified in 3.5.5.2 by alternately switching S2 between positions 1 and 2. Using oscilloscope, item 5, observe delays from TP21 to TP4 as specified in 3.5.5.2.

4.7 Soldering. As specified in MIL-C-50652.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.



*REFER TO 4.5.3

FIGURE 5. RCFD DECODER TEST SETUP.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing, and marking shall be in accordance with Packaging Data Sheet PS10548119. The level of protection shall be as specified in the procurement document.

6. NOTES. As specified in para 6.0 of MIL-C-50652.

6.1 Changes from Previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-Ar

Preparing activity:
Army-AR

(Project 1285-A878)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-50652/3A	09 April 1993

3. DOCUMENT TITLE: CIRCUIT CARD ASSEMBLY: 10548119 (DECODER, RCFD)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Barbara Irwin	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) U.S. Army ARDEC (SMCAR-[illegible]) Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: [illegible] and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3464 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:31Z
Check the source to verify that this is the current version before use.